

Replication of Insider Trading and Innovation by *R. Levine et al.*

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1 Introduction

The paper examines whether legal systems that restrict insider trading, (i.e. trading by corporate officials or major shareholders on material nonpublic information) influence the invention of new technologies, as measured by patent intensity, scope, impact, generality, and originality. The ratio behind this study builds on DeMarzo, Fishman, and Hagerty (1998), which shows that insider trading allows corporate insiders to exploit other investors, thus discouraging investment, and therefore, innovation. Moreover, innovative activities are particularly subject to the lack of trust, because generally more difficult to assess than more mainstream activities (Holmstrom 1989).

1.1 Dataset

For a set of countries, the year when each country first enacts (i.e. passes) insider trading laws and enforces (i.e. when the first violator is prosecuted) such laws is obtained from Bhattacharya and Daouk (2002). According to Bhattacharya and Daouk (2002), enforcement is more relevant than just enactment. The measures for innovations are 6 patent-based indicators, using industry-level data for each country, ranging between year 1976 and 2006, obtained from the European Patent Office's Worldwide Patent Statistical Database (PAT-STAT). This dataset consists of 83200 industry-country-year observations. The indicators are:

- Patent Count: number of patents
- Citation Weight: number of forward citations to patent filed
- Top 10% Patents: number of patents that become part of the top 10% citation in their technology class
- Patent Entities: number of patenting entities

- General Innovation: degree of citations of the patents from other technology classes patents
- Originality: degree of which the patents cite patents in other technology classes

2 Methodology

2.1 Preliminary analysis

For a first analysis, the data is aggregated to a country level. The baseline regression equation is thus:

$$Y_{ict} = \alpha_0 + \alpha_1 E_{ct} + \gamma X'_{ict} + \delta_c + \delta_i + \delta_t + \varepsilon_{ict}, \quad (1)$$

where the indices (i, c, t) refer to, respectively, industry, country and year; Y_{ict} is one of the six patent-based indicator, the δ 's are country, industry and time fixed effects; X_{ict} is a vector of controls, such as: $\log GDP$, $\log GDP/Population$, $Stock/GDP$, $Credit/GDP$, $Trade$, all obtained from the World Development Indicators and the Financial Development Infrastructure of the World Bank; and finally E_{ct} is the insider trading law enforcement dummy: for each country, it is 0 if there are no insider trading laws enforced, and 1 otherwise. The coefficient of interest for the treatment effect is thus α_1 . The authors thus perform the regression using the Regression with Double Fixed Effects and Heteroskedasticity-Robust Standard Errors Stata package `reghdfe` and find that the enforcement of insider-trading laws is associated with a statistically significant, economically large and highly robust increase in each of the six patent-based indicators, ranging from 15% to 29%, as shown in table 1.

2.2 Industry level analysis

The authors then proceed to evaluate the problem at industry granularity: they introduce an industry dummy I_i to the baseline regression:

$$Y_{ict} = \beta_0 + \beta_1 I_i E_{ct} + \lambda X'_{ict} + \delta_{ct} + \delta_{it} + \varepsilon_{ict}, \quad (2)$$

finding similar results to the baseline regression.

2.3 Robustness and validation

The authors perform various tests and robustness checks, including:

- Check for omitted variable bias by including and excluding a large set of additional controls, composed of several indicators of securities-market reforms and policies toward international capital flows; an array of indicators of bank regulatory and supervisory policies; and enactment of patent laws, measures of intellectual-property-rights protection, property-rights protection more generally, and the effectiveness of the legal system. They find that including or excluding these controls do not alter the results.

Table 1: Author's baseline estimates, without and with controls

Enforcement and Innovation: Baseline Results						
	Patent Count	Patent Entities	Citation	PC Top 10 Percent	Generality	Originality
Enforce	.2118*	.1661*	.3623**	.1338**	.1214**	.2235**
	(2.68)	(2.12)	(2.84)	(3.71)	(3.27)	(4.09)
Adjusted R ²	.839	.848	.847	.719	.762	.760
Enforce	.1547*	.1186*	.2948*	.1094**	.1059**	.1679**
	(2.44)	(1.94)	(2.59)	(3.50)	(3.43)	(3.58)
Control variables:						
Enact	.0098	.0109	-.0262	-.0022	.0034	-.0174
	(.25)	(.28)	(-.51)	(-.19)	(.25)	(-.94)
GDP	-.0011	.0525	-.1834	-.2136	-.1061	-.4966*
	(-.00)	(.21)	(-.37)	(-1.63)	(-.80)	(-2.66)
GDP per Capita	.4602*	.4593*	1.1689**	.3020*	.3436**	.6895**
	(2.29)	(2.40)	(2.95)	(2.71)	(3.20)	(4.28)
Stock/GDP	.1225*	.0970*	.2347*	.0791**	.0643*	.1097**
	(2.15)	(1.73)	(2.22)	(2.76)	(1.91)	(2.78)
Credit/GDP	.1155	.0995	.2213	.0236	.0210	.0501
	(1.58)	(1.34)	(1.60)	(.48)	(.46)	(.88)
Export to United States	1.2157**	1.0647**	1.4617**	.8264**	.9999**	.9820**
	(6.21)	(6.58)	(6.20)	(4.69)	(5.16)	(5.26)
Adjusted R ²	.853	.862	.861	.728	.771	.773

Note. Values are baseline effects of initial enforcement on innovation measured at the industry-country level using equation (1). The dependent variable is the natural logarithm of 1 plus the raw value of innovation measures at the industry-country-year level. All regressions include country, industry, and year fixed effects. Robust *t*-statistics based on standard errors clustered at the country-year level are in parentheses. $N = 83,200$ for regressions without control variables; $N = 76,561$ for regressions with control variables.

* $p < .10$.

* $p < .05$.

** $p < .01$.

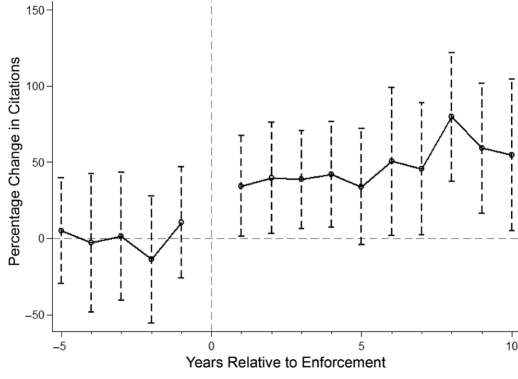
- Do not find any pretrends in the patent-based measures of innovation before enforcement, based on graphical analysis.
- Find that the growth rates in innovation measures do not predict the enforcement of the laws, using a Hazard-Model estimation.

2.4 Dynamic

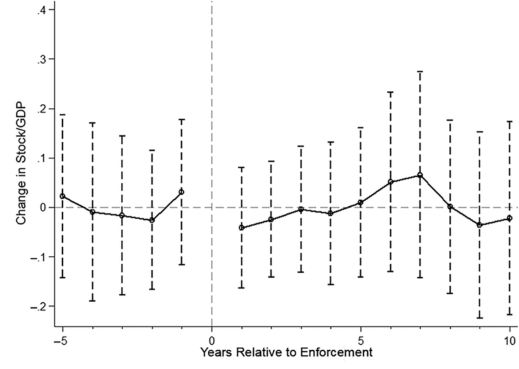
The authors study the behaviour of the innovation proxies in a $(-5, 10)$ window around the enforcement year, implementing:

$$Y_{ct} = \alpha_0 + \sum_{\tau=-5}^{10} \alpha_{1,\tau} E_{c,t,\tau} + \delta_c + \delta_t + \varepsilon_{ct} \quad (3)$$

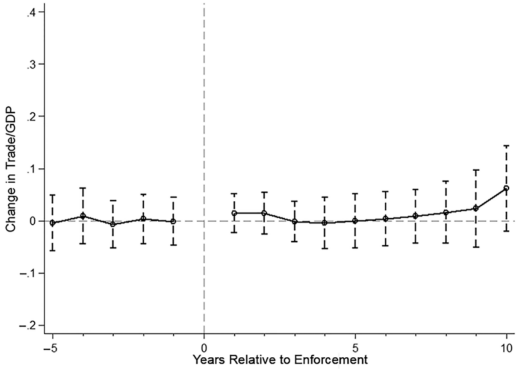
They find a sizeable increase in innovation around the treatment year, while the control variables stay almost constant, as can be seen in figure 1.



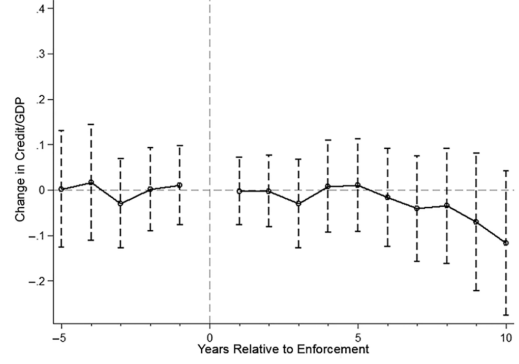
(a) Citation Weight



(b) Stock Market Capitalization



(c) Trade



(d) Credit

Figure 1: Dynamic Effects on Citation Weight, Stock Market Capitalization, Trade and Credit

3 Extension

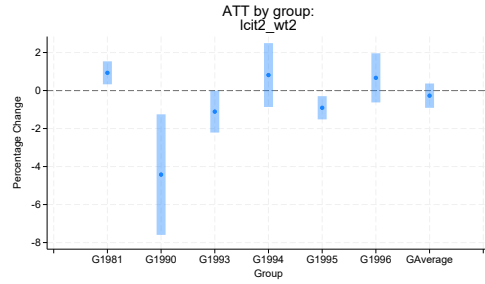
The approach taken by the authors for both the baseline and the industry-level study may be considered naive and outdated: as previously treated units are included in the subsequent control groups, the resulting estimate for the α_1 's and β_1 's is a weighted average of the treatment effects for each group, with weights that may even be negative, and thus potentially biased. To address this issue, I adopt a modern diff-in-diff technique proposed by Callaway and Sant'Anna (2021), implemented in Stata in the `csdid` package. It offers two primary advantages:

1. **Enhanced Control Group Selection:** It provides flexibility in choosing control groups. Specifically, it allows to utilize *not-yet treated units* as controls in addition to *never-treated* units (as this dataset lacks never treated groups).
2. **Covariate Incorporation:** it accommodates the inclusion of covariates, thereby ensuring that the parallel conditions assumption holds conditionally on these covariates, and implicitly handling the fixed effects.

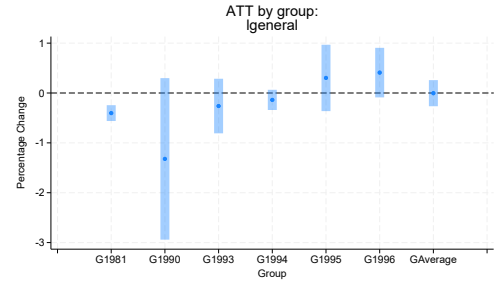
In particular, I focus on evaluating the average treatment effect (ATT) by treatment group, reported in 2. Moreover, I reproduce, in figure 3 the dynamics in the $(-5, 10)$ year window around the enforcement year.

3.1 Treatment Effect by Group

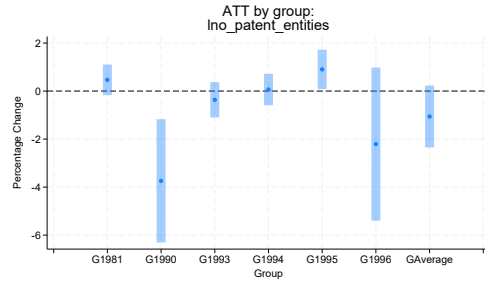
Table 2 and its graphical representation in figure 2 reproduce the average treatment effect for different groups, using the industry-level dataset. A similar analysis has been performed on the country level dataset, and in both cases, the treatment effect are found to be not statistically relevant.



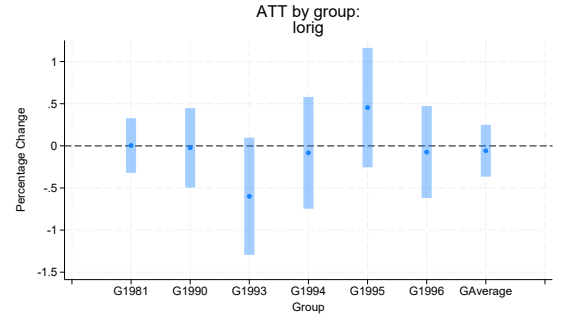
(a) Citation Weight



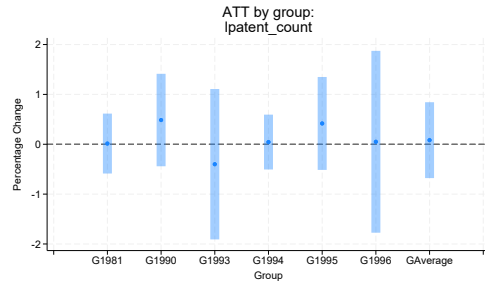
(b) General Innovation



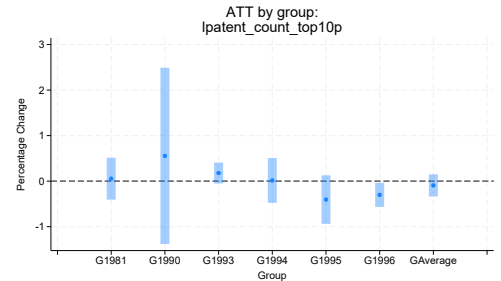
(c) Patent Entities



(d) Originality



(e) Patent Count



(f) Top 10% Patents

Figure 2: ATT Estimates by Group Across Innovation Measures

Table 2: ATT by Group

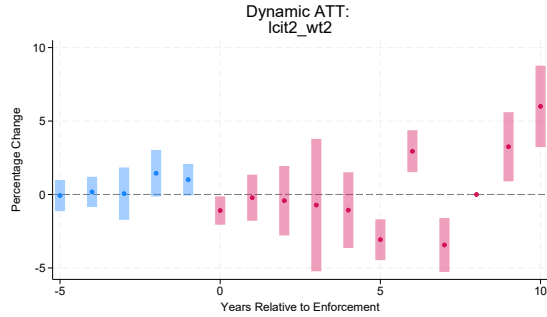
	(1)	(2)	(3)	(4)	(5)	(6)
	Patent Count	Patent Entities	Citation Weight	Top 10% Patents	General Innovation	Originality
GAverage	0.0823 (0.388)	-1.059 (0.657)	-0.270 (0.326)	-0.0946 (0.124)	-0.00366 (0.134)	-0.0588 (0.157)
G1981	0.0141 (0.307)	0.469 (0.324)	0.933*** (0.308)	0.0542 (0.235)	-0.403*** (0.0810)	0.00337 (0.165)
G1990	0.486 (0.473)	-3.740*** (1.310)	-4.424*** (1.618)	0.555 (0.987)	-1.322 (0.826)	-0.0240 (0.241)
G1993	-0.400 (0.770)	-0.363 (0.376)	-1.111** (0.562)	0.179 (0.117)	-0.262 (0.279)	-0.599* (0.355)
G1994	0.0439 (0.281)	0.0655 (0.335)	0.814 (0.857)	0.0152 (0.251)	-0.138 (0.103)	-0.0832 (0.338)
G1995	0.417 (0.476)	0.904** (0.417)	-0.908*** (0.311)	-0.405 (0.272)	0.302 (0.340)	0.454 (0.362)
G1996	0.0498 (0.930)	-2.207 (1.626)	0.668 (0.659)	-0.302** (0.135)	0.409 (0.254)	-0.0736 (0.278)
Observations						

Standard errors in parentheses

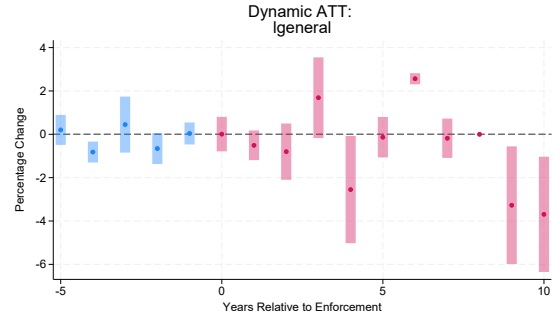
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

3.2 Dynamics

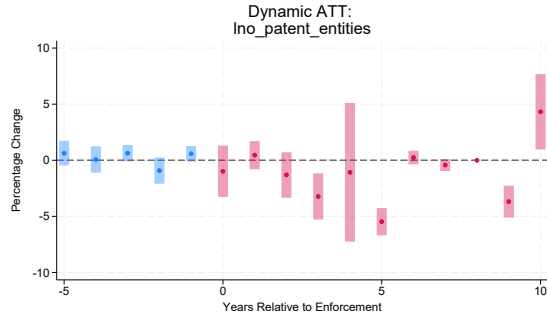
As reported in the following figure 3 and table 3, also in the event aggregation of the $ATT(g, t)$ functions the effects are mostly statistically insignificant.



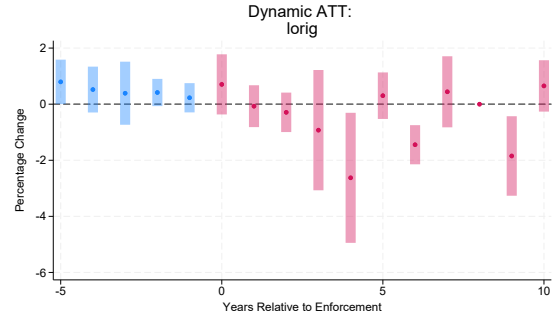
(a) Citation Weight



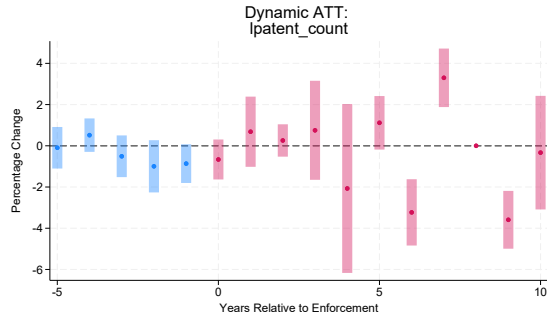
(b) General Innovation



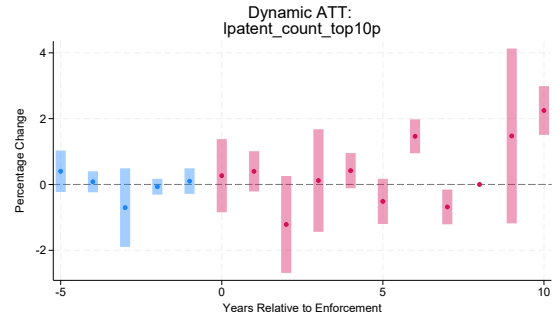
(c) Patent Entities



(d) Originality



(e) Patent Count



(f) Top 10% Patents

Figure 3: Dynamic ATT Estimates Across Innovation Measures

Table 3: Dynamic ATT by Event Time

	(1)	(2)	(3)	(4)	(5)	(6)	
	Patent Count	Patent Entities	Citation Weight	Top 10% Patents	General Innovation	Originality	
∞	Pre_avg	-0.389 (0.300)	0.199 (0.260)	0.523** (0.238)	-0.0369 (0.113)	-0.157 (0.244)	0.470** (0.188)
	Post_avg	0 (.)	0 (.)	0 (.)	0 (.)	0 (.)	0 (.)
	Tm5	-0.0947 (0.514)	0.635 (0.562)	-0.0709 (0.542)	0.402 (0.321)	0.203 (0.353)	0.795** (0.404)
	Tm4	0.517 (0.414)	0.0696 (0.598)	0.176 (0.526)	0.0820 (0.163)	-0.818*** (0.246)	0.521 (0.417)
	Tm3	-0.508 (0.516)	0.631* (0.374)	0.0577 (0.910)	-0.703 (0.610)	0.447 (0.659)	0.391 (0.574)
	Tm2	-0.995 (0.646)	-0.918 (0.598)	1.447* (0.809)	-0.0681 (0.122)	-0.657* (0.365)	0.416* (0.248)
	Tm1	-0.864* (0.479)	0.576 (0.351)	1.007* (0.549)	0.102 (0.198)	0.0392 (0.259)	0.228 (0.266)
	Tp0	-0.664 (0.495)	-0.982 (1.166)	-1.093** (0.491)	0.267 (0.567)	0.00942 (0.406)	0.706 (0.547)
	Tp1	0.683 (0.869)	0.459 (0.639)	-0.218 (0.802)	0.400 (0.312)	-0.507 (0.349)	-0.0711 (0.381)
	Tp2	0.258 (0.401)	-1.308 (1.034)	-0.423 (1.206)	-1.215 (0.751)	-0.800 (0.663)	-0.290 (0.359)

Tp3	0.752 (1.226)	-3.226*** (1.049)	-0.720 (2.299)	0.120 (0.795)	1.688* (0.950)	-0.926 (1.094)
Tp4	-2.070 (2.091)	-1.072 (3.154)	-1.064 (1.314)	0.421 (0.272)	-2.550** (1.262)	-2.626** (1.183)
Tp5	1.115* (0.663)	-5.474*** (0.620)	-3.080*** (0.709)	-0.514 (0.350)	-0.133 (0.476)	0.302 (0.423)
Tp6	-3.230*** (0.821)	0.241 (0.311)	2.945*** (0.728)	1.464*** (0.263)	2.563*** (0.130)	-1.446*** (0.356)
Tp7	3.299*** (0.724)	-0.423 (0.272)	-3.436*** (0.938)	-0.682** (0.269)	-0.181 (0.463)	0.441 (0.647)
Tp8	0 (.)	0 (.)	0 (.)	0 (.)	0 (.)	0 (.)
Tp9	-3.590*** (0.715)	-3.686*** (0.725)	3.249*** (1.201)	1.474 (1.355)	-3.271** (1.385)	-1.847** (0.723)
Tp10	-0.333 (1.406)	4.320** (1.715)	6.000*** (1.414)	2.248*** (0.378)	-3.692*** (1.357)	0.651 (0.467)

Observations

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4 Conclusion

The results of the paper do not seem to hold when confronted with more advanced DiD techniques, and the `csdid`-estimated ATT are mostly not statistically significant, both when aggregated by group and in the overall average. Therefore, we can conclude that enforcing insider trading laws alone is unlikely to have the large effect claimed by the authors, and other confounding effects may be at play.

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